

Welly Oktariana, S.T

Data Analyst | Data Scientist

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Portfolio: wellyokt.github.io/portfolio-welly/

SUMMARY

Data Analyst with 3+ years of experience creating data visualizations, performing spatial analysis, and building machine learning models that support business decisions. Skilled at combining spatial and non-spatial data across both large enterprise-scale datasets and smaller exploratory analyses to uncover patterns, identify trends, and translate complex information into clear business recommendations.

PROFESSIONAL EXPERIENCE

PT Bhumi Varta Technology — Data Analyst

May 2023 - present

- developed sales and site-selection prediction models across multiple enterprise clients using demographic and POI data, achieving up to **90% predictive accuracy**, including cannibalization analysis to support expansion decisions.
- Researched and built a data quality assessment pipeline, combining spatial validation rules with metadata analytics, achieving **90% precision** in flagging valid data and reducing manual validation effort
- Built an ETL pipeline aggregating and transforming POI, road, and administrative boundary data into a nationwide H3-based location index, refreshed on a 6-month cycle to track area growth trends, with results delivered through an interactive dashboard.
- Analyzed large-scale mobility data — including dwell time, home origin, and interest/behavioral patterns — to derive location-based audience insights for site and market decisions.
- Built a site evaluation platform combining an AI-powered chatbot (LangChain, LangGraph) for interactive Q&A on analysis results, with findings visualized through an AI-enhanced dashboard for stakeholder use.
- Researched derivative data and new methodologies to support and strengthen analysis quality.
- Built interactive dashboards and analytical applications for reporting; presented findings directly to stakeholders.

PT Bhumi Varta Technology — GIS Technician

Oct 2021 - Apr 2023

- Led preparation, cleaning, and structuring of Point-of-Interest (POI) datasets.
- Defined and implemented data validation methodologies to maintain data quality and consistency across projects
- Managed a team of 10 freelance data operators, overseeing daily data collection and input tasks.
- Built Excel-based tracking tools to monitor freelancer output, task progress, and team performance.
- Conducted quality control on freelancer output, reviewing and verifying data accuracy before final delivery.

Badan Pendapatan Daerah DKI Jakarta — Mapping Surveyor

- Conducted field surveys to validate PBB-P2 (land and building tax) map data in Palmerah.
- Processed survey data, including polygon editing and attribute data input.
- Reported progress to Field and Regional Managers.

PERSONAL PROJECT

Palm Tree Detection from Aerial Imagery (Computer Vision)

Self-initiated

- Trained a YOLO11n object detection model to detect and count oil palm trees from aerial imagery, applying computer vision techniques to a real-world agricultural use case.

House Price Prediction with Computer Vision

Final Project — Dibimbing.id AI/ML Program

- Built a combined prediction model integrating object detection results from satellite imagery with tabular features (number of rooms, land size, building size) to estimate house prices.

EDUCATION

Institut Teknologi Sumatera — Bachelor of Geomatics Engineering

2016 - 2020

GPA 3.17/4.00 | Thesis: The Effect of Tidal Variations on Coastline Delineation and Maritime Boundary Determination

PROFESSIONAL TRAINING

Dibimbing.id — AI Machine Learning

Oct 2024 - Mar 2025

- Machine Learning & AI: supervised/unsupervised learning, feature engineering, hyperparameter tuning.
- Advanced techniques: ensemble learning, time series modeling, PCA, RFM analysis.
- Deep learning: Computer Vision, NLP, OCR, recommendation systems.

SKILLS

Programming: SQL, Python (Pandas, NumPy, GeoPandas)

Database: PostgreSQL, Databricks Unity Catalog

Machine Learning: Scikit-Learn, TensorFlow, Regression, Classification, Clustering, Feature Engineering, Hyperparameter Tuning

LLM: LangChain, LangGraph, SQLite

Cloud & Tools: Microsoft Azure, Azure Databricks, Azure Blob Storage, Azure Repos, Git

Visualization & Reporting: Tableau, Power BI, Streamlit, Matplotlib, Seaborn, Plotly, Excel, Google Sheets, HTML Reports